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# Module 1
# Introduction

## Introduction to RAG

# Module Overview

**Introduction to RAG**

**RAG Architecture Deep Dive**

**Real-World Applications**

# Hands-on Projects



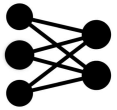
**Module 1**
Basic RAG



**Module 2**
Retriever



**Module 3**
Vector Database



**Module 4**
LLM



**Module 5**
Monitoring & Evaluation

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# Introduction to RAG

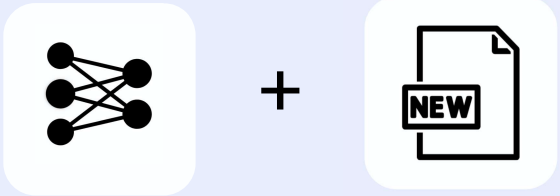
## Introduction to RAG

# Retrieval Augmented Generation (RAG)

**LLMS are already powerful**

- Summarize Text
- Generate Code
- Rewrite Content

**RAG further improves them**




LLM + New information

# Why are hotels expensive on the weekend?

"More people travel on weekends, so there’s more competition for rooms.”

# Why are hotels in Vancouver super expensive this coming weekend?

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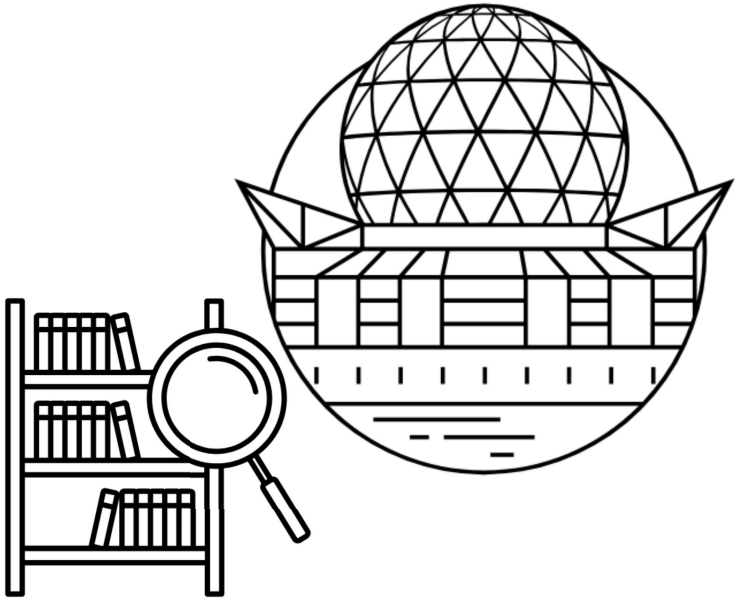
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# Why are hotels in Vancouver super expensive this coming weekend?

Taylor Swift is in town this weekend



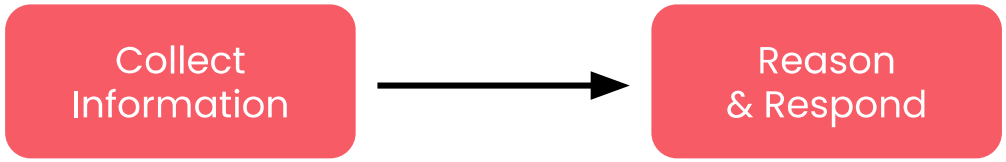
# Why doesn’t Vancouver have more hotel capacity close to downtown?



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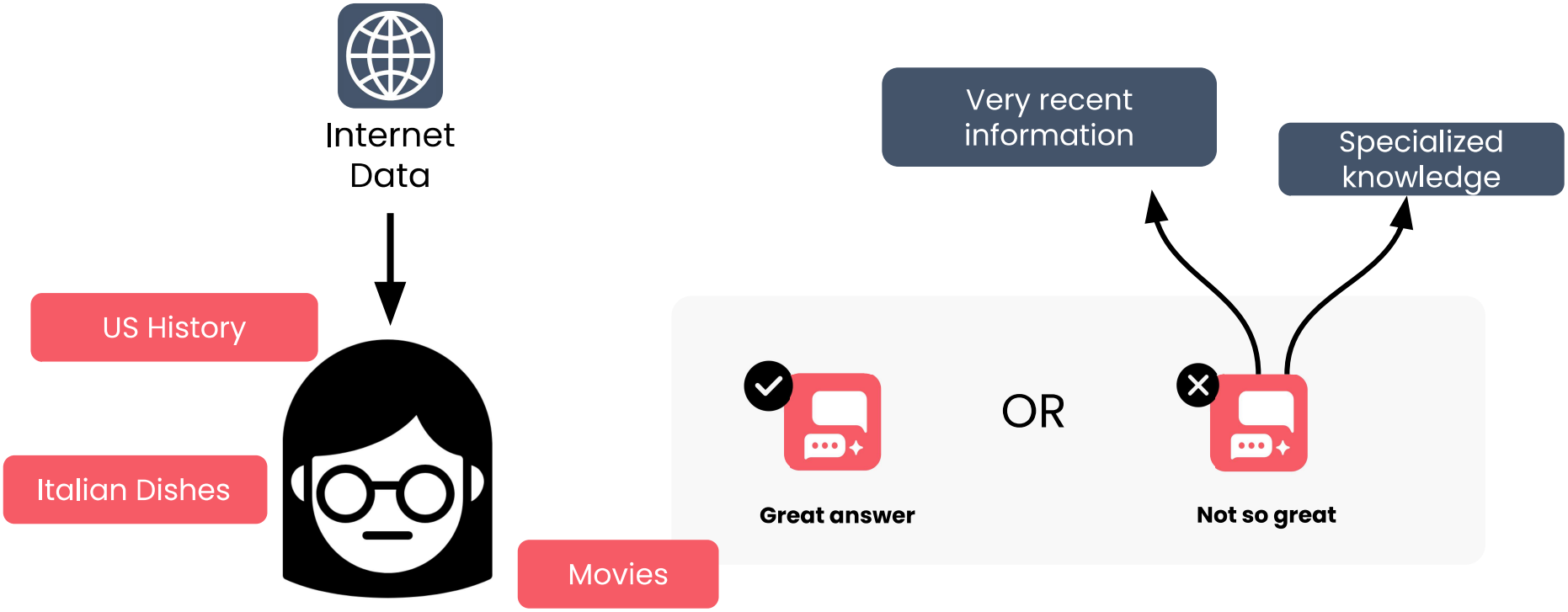
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# Two Steps for Answering Questions

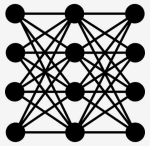



| | | |
|---|---|---|
| **Why are hotels expensive on the weekend?** | No need to collect information | Respond based on your knowledge |
| **Why doesn't Vancouver have more hotel capacity close to downtown?** | Extensive Research | Synthesize research |

# Traditional Language Model Usage



LLM



Mathematical Models

**User Query**

Why doesn't Vancouver have more hotel capacity close to downtown?



**Urban Development in Vancouver**
Zoning laws and city planning since the early 1900s limited hotel growth downtown.



**Forum Comment**
Probably because land costs are super high near downtown.

Training Data

**Massive dataset from the open internet**

# What LLMs don’t know

**Private Databases**
LLMS can’t access confidential information

**Hard to access information**
Some information isn’t widely available online, making it inaccessible to LLMs.

**Real time data**
LLMs are trained on past data and don’t update automatically

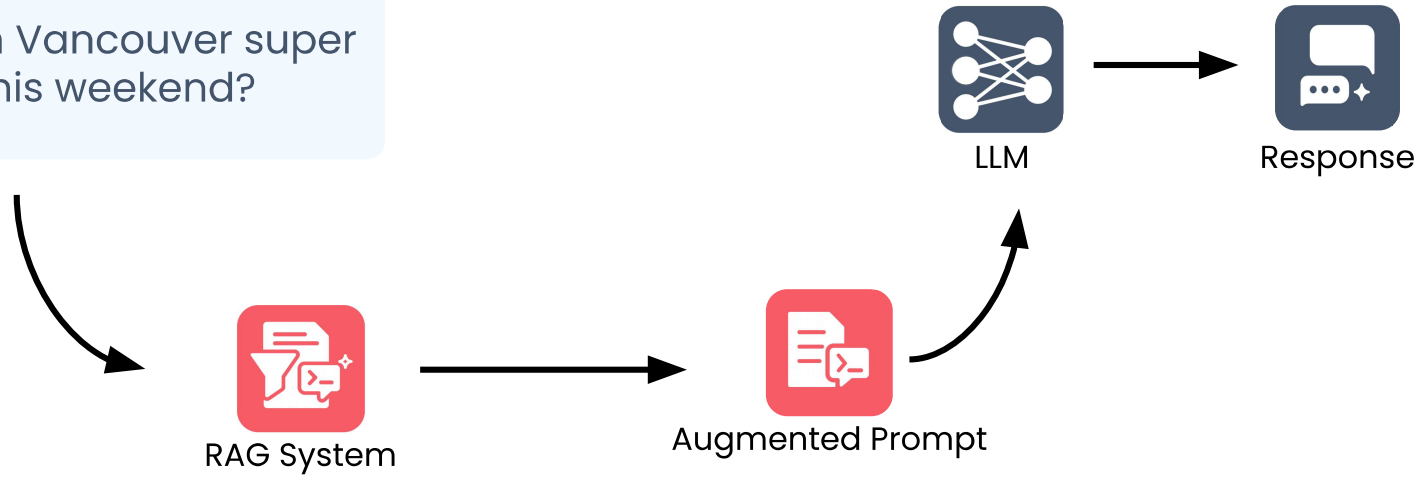
**How do you make sure the LLM knows this useful information?**

# Just put it in the prompt!



**User Prompt**

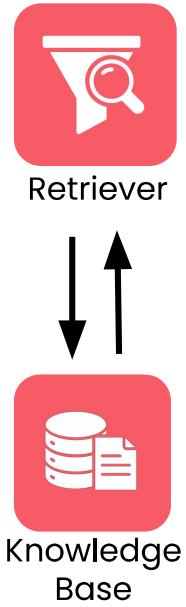
Why are hotels in Vancouver super expensive this weekend?




```
Taylor Swift is performing her Eras Tour in
Vancouver. This weekend at BC Place Stadium on
December 6-8, 2024
```

# Retriever

- Manages knowledge base of trusted information
- Finds the most relevant Information and shares information with the LLM
- Improves generation



# Retrieval Augmented Generation

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# Applications of RAG

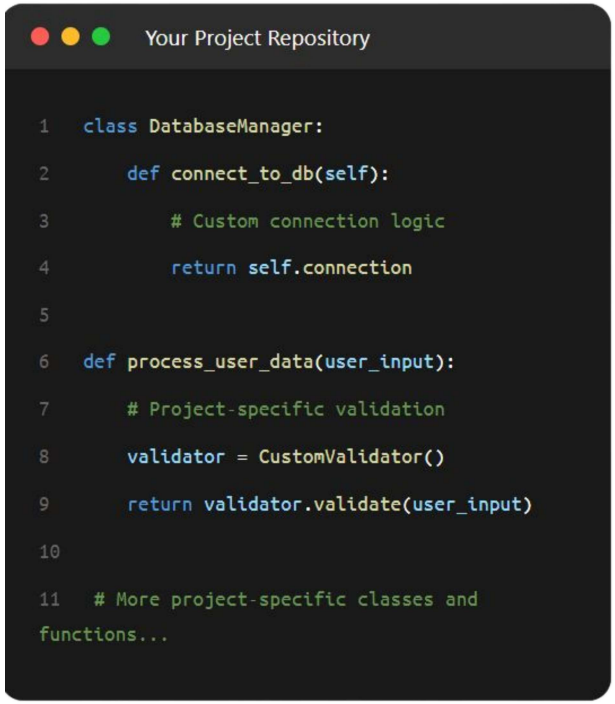
## Applications of RAG

# Applications of RAG - Code Generation

**LLM needs your project's context**
Classes, functions, definitions, and coding style

**Use your codebase as a knowledge base**
RAG retrieves project-specific content for the LLM

**Improves code generation and Q&A**
Answers are tailored to your actual repository

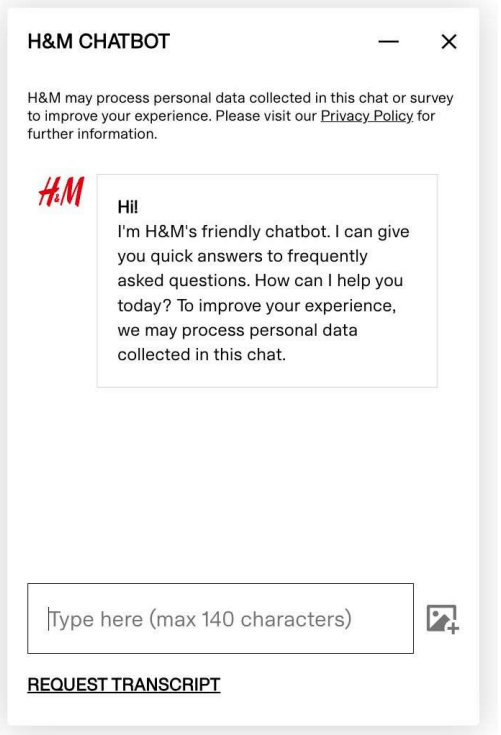


# Applications of RAG - Company Chatbots

**Tailored to your company**
Every business has its own products, policies, and communication style

**Uses your internal documents**
Manuals, support guides, FAQs

**Grounds answers in real context**
Reduces generic or incorrect answers



# Applications of RAG - Specialized Knowledge

**High-impact domains**
Legal and medical use cases

**Uses specialized documents**
Case files, journals, private data

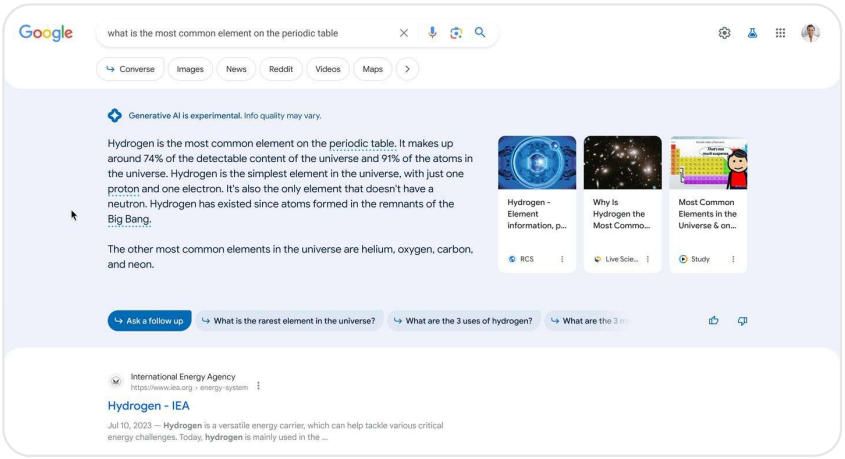
**Enables accurate, secure use**
Supports precision and privacy needs

# Applications of RAG

**Search engines as retrievers**
Return websites for a given query

**AI summarizes search results**
Presents key info in a skimmable format

**RAG with the internet as a knowledge base**
Summaries powered by real-time retrieval




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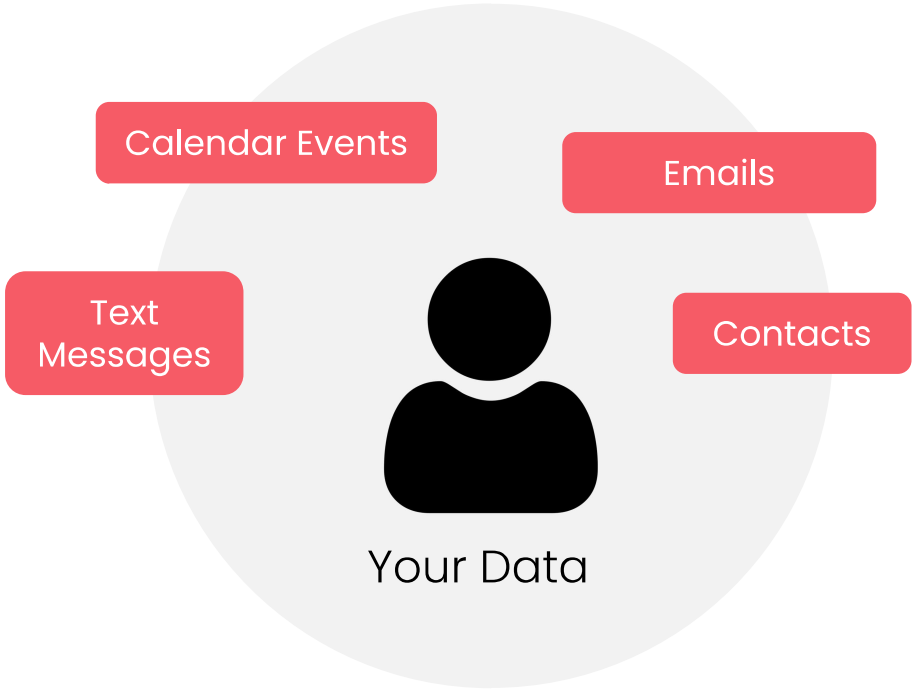
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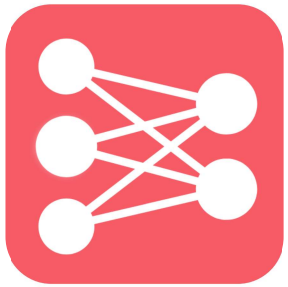
# Personalized RAG

**More software includes personal assistants**
Messaging app, email client, etc.

**These tools need context**
More context leads to better results

**Your data is the knowledge base**
Texts, contacts, etc.





+ your data

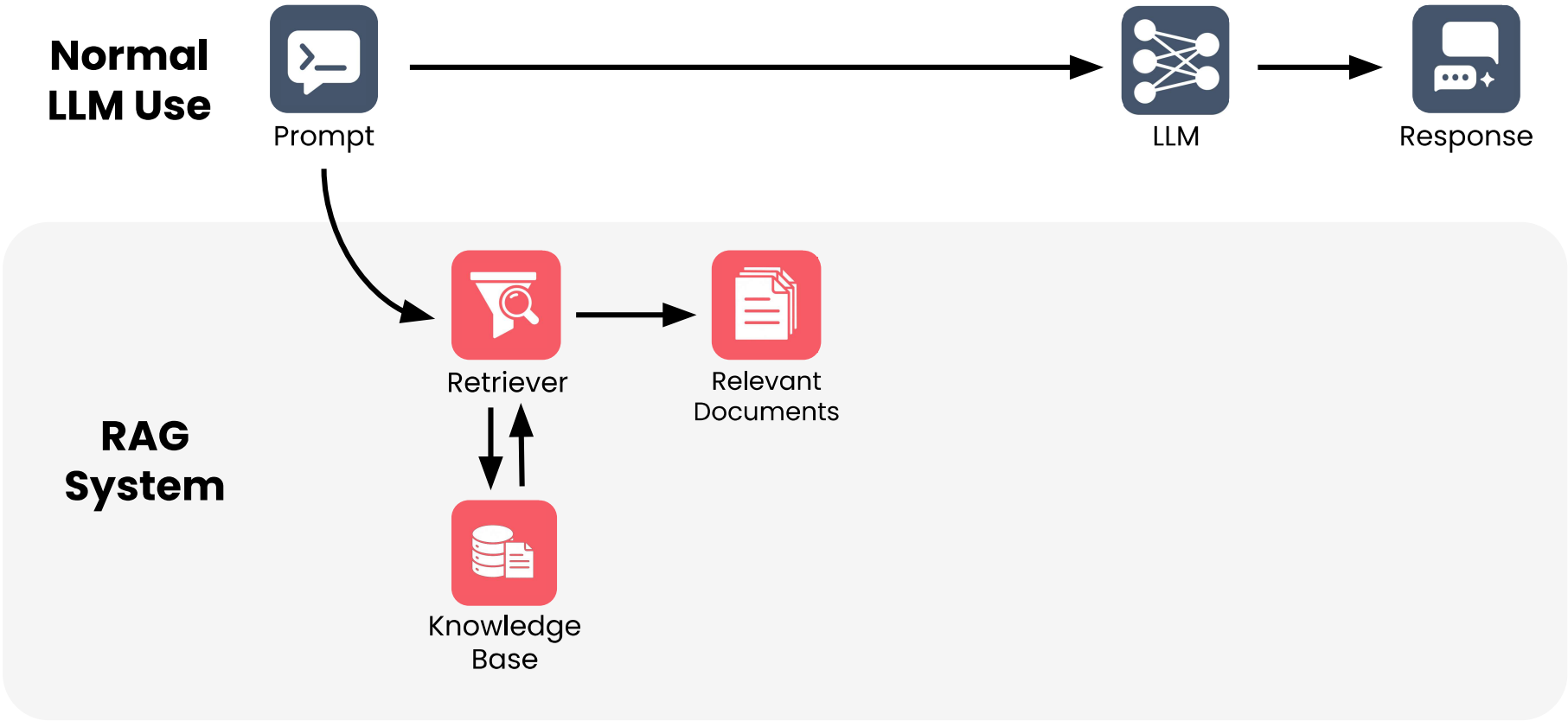
LLM

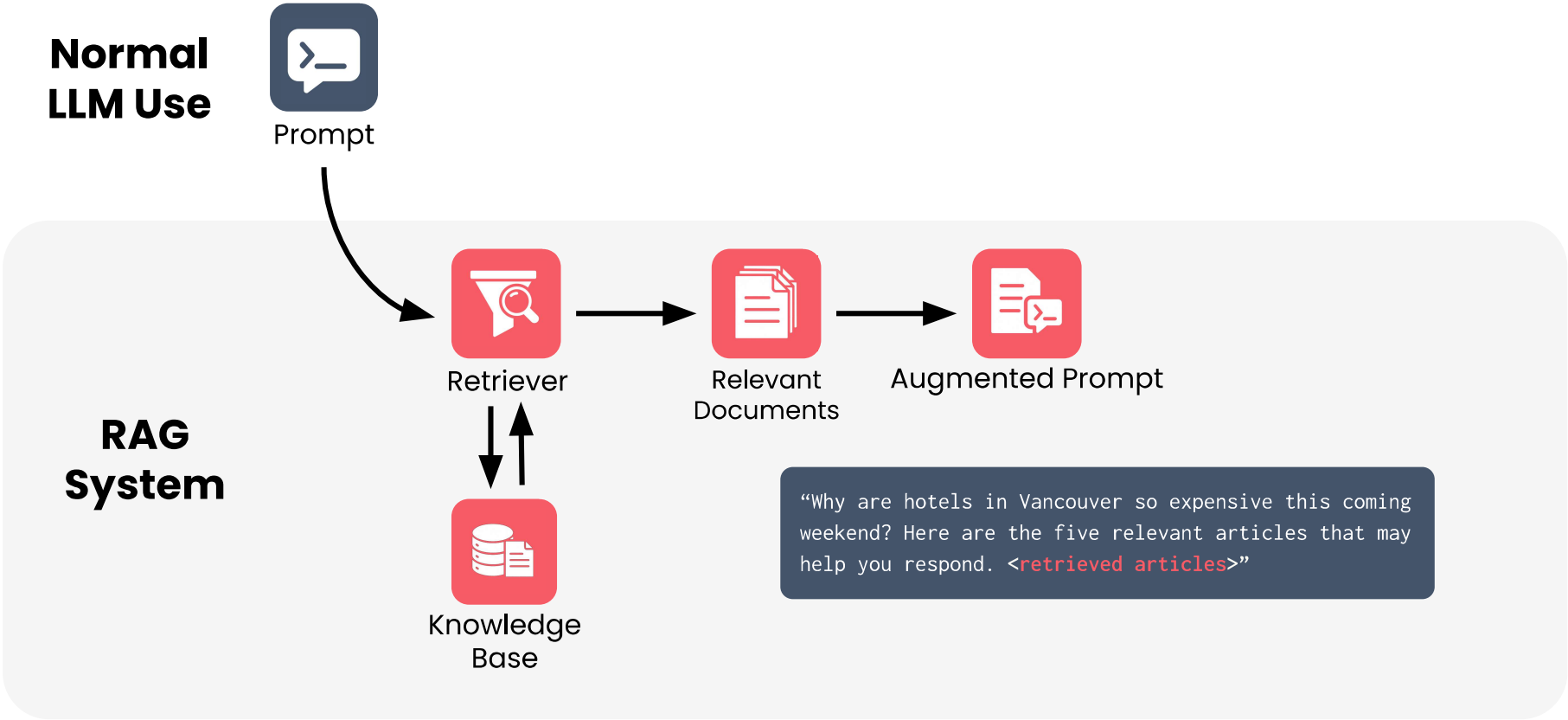


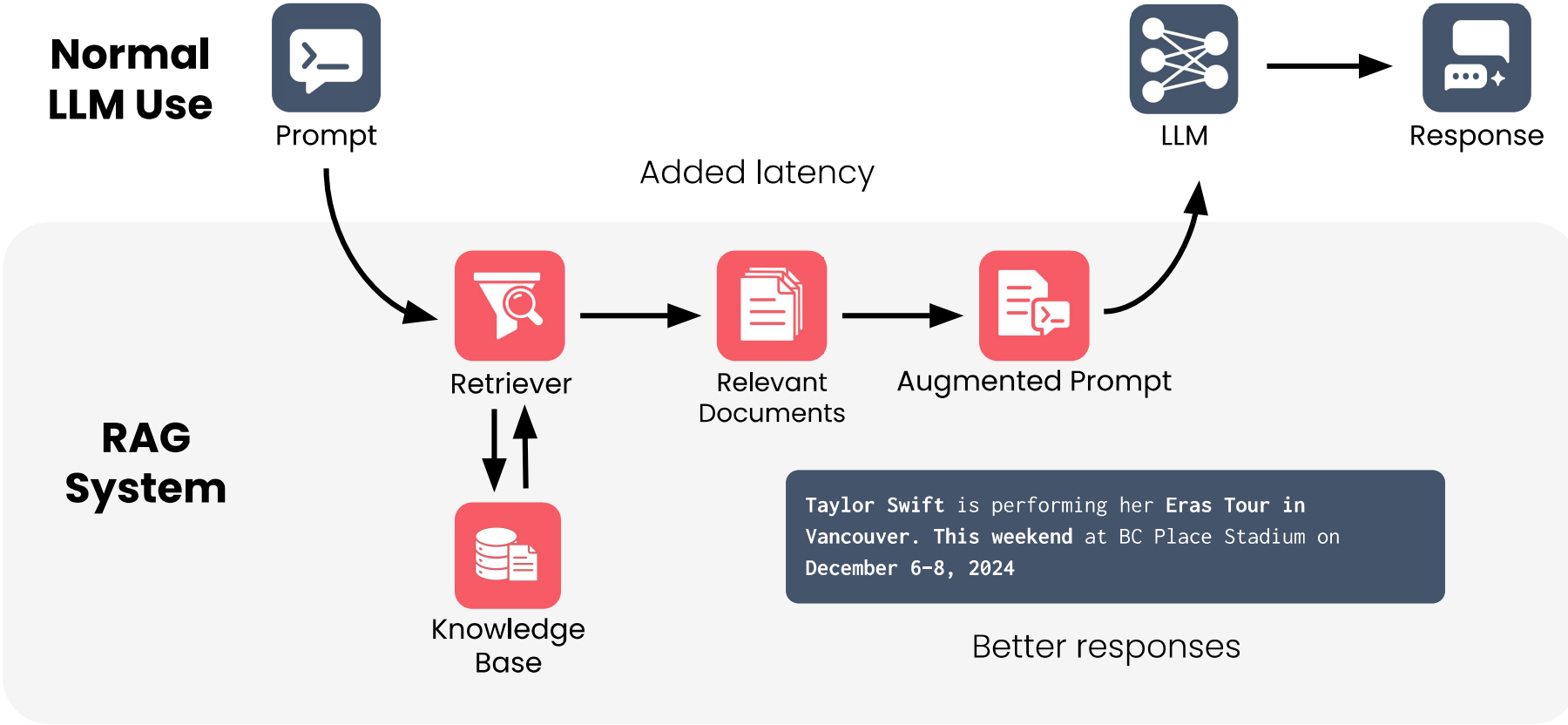
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# RAG Architecture Overview

## Introduction to RAG


Normal
LLM Use
Prompt
LLM
Response
RAG
System
Retriever
Relevant
Documents
Knowledge
Base


Normal
LLM Use
Prompt
RAG
System
Retriever
Relevant
Documents
Augmented Prompt
Knowledge
Base
“Why are hotels in Vancouver so expensive this coming weekend? Here are the five relevant articles that may help you respond. <retrieved articles>”


Normal LLM Use
Prompt
LLM
Response
Added latency
RAG System
Retriever
Relevant Documents
Augmented Prompt
Knowledge Base
Taylor Swift is performing her Eras Tour in Vancouver. This weekend at BC Place Stadium on December 6-8, 2024
Better responses

# Advantages of RAG

**Injects missing knowledge**
Adds info not in the training data (e.g. policies, updates)

**Reduces hallucinations**
Grounds answers with relevant context

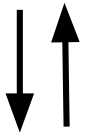
**Keeps models up to date**
Reflects new info by updating the knowledge base

**Enables source citation**
Includes sources for verifiable answers

**Focuses model on generation**
Retriever finds facts, LLM writes responses



Retriever





Knowledge Base

```
“Why are hotels in Vancouver so expensive this coming weekend? Here are the five relevant articles that may help you respond. <retrieved articles>”
```



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# Introduction to LLMs

## Introduction to RAG

LLMs are just fancy autocomplete

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What a beautiful day, the sun is **out** **Prompt**

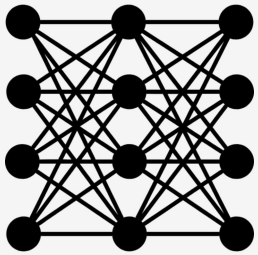
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What a beautiful day the sun is shining

What a beautiful day the sun is rising **Completions**

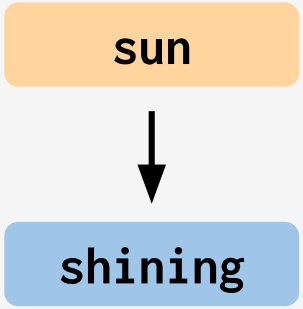
What a beautiful day the sun is out

What a beautiful day, the sun is `exploding`



**Neural Network**

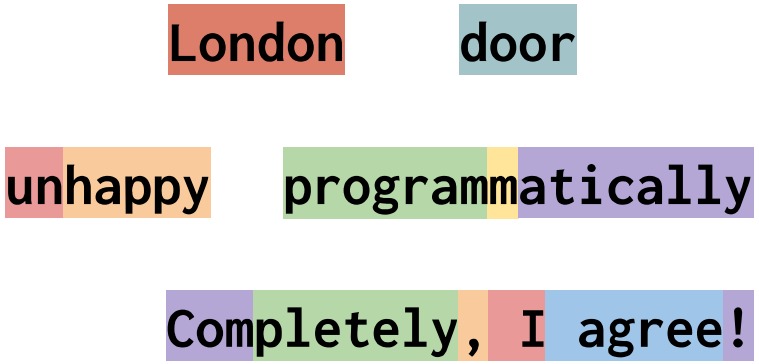
- a complex mathematical model of language
- stores which words frequently appear together, in which order, and contextual meaning
- LLMs use this model to generate text



What a beautiful day, the sun is shining in the sky

## Token

- a piece of a word
- some words get single tokens
- compound words use multiple tokens
- punctuation marks
- ~10,000 - 100,000 tokens in LLM's vocabulary, allowing models to represent any possible word with fewer tokens

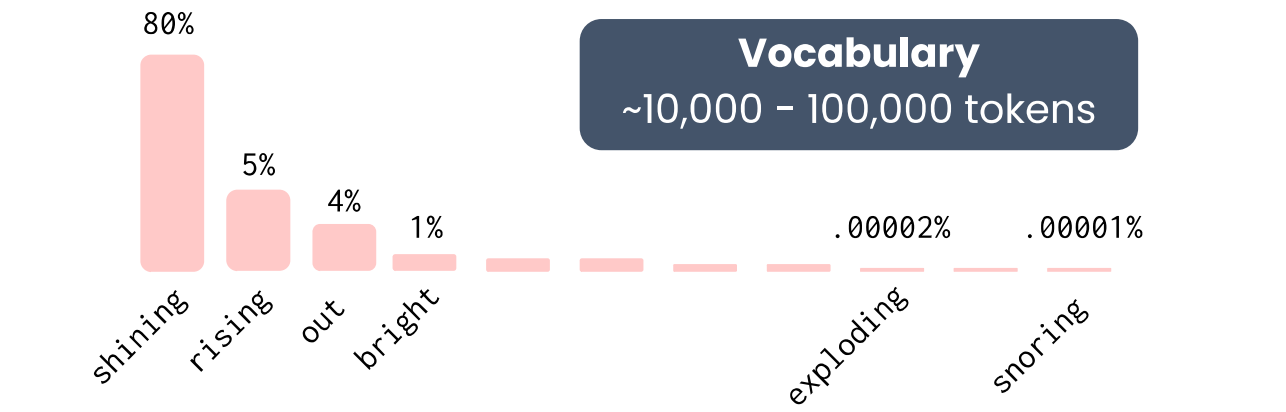


What a beautiful day, the sun is

Process Current State

Calculate Probabilities

Select Next Token




**shining**

What a beautiful day, the sun is **shining**

Process Current State

Calculate Probabilities

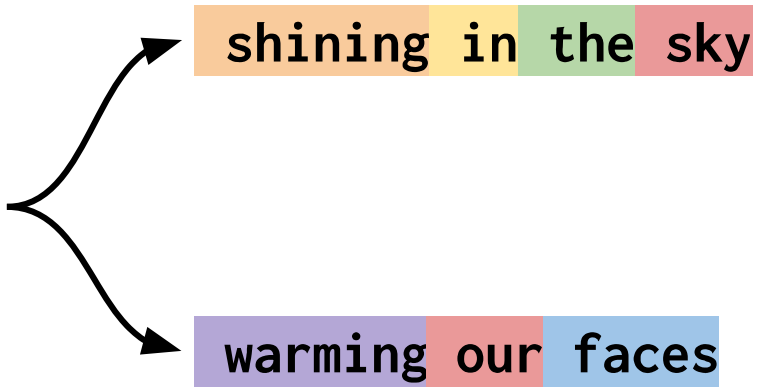
Select Next Token




new tokens make sense in context of old ones

**in**

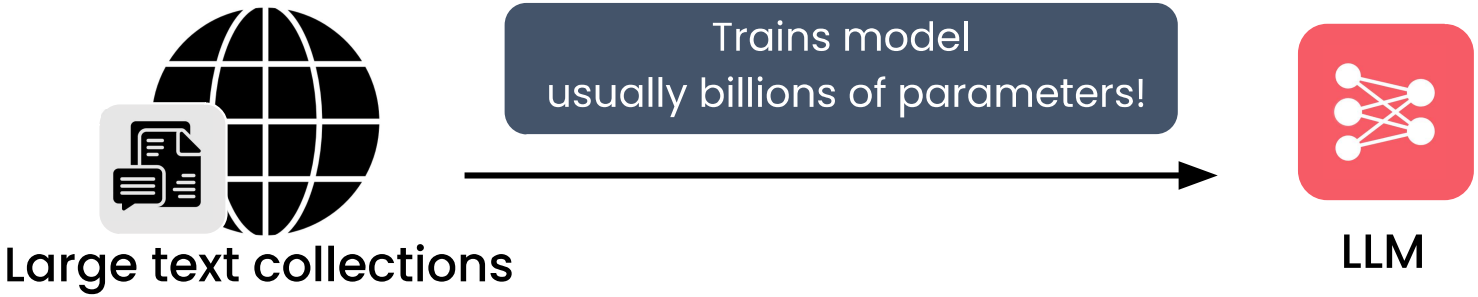
What a beautiful day, the sun is




**Autoregressive**

- “self-influencing”
- new tokens make sense in context of old ones
- running the same prompt leads to different completion

# How LLMs Learn

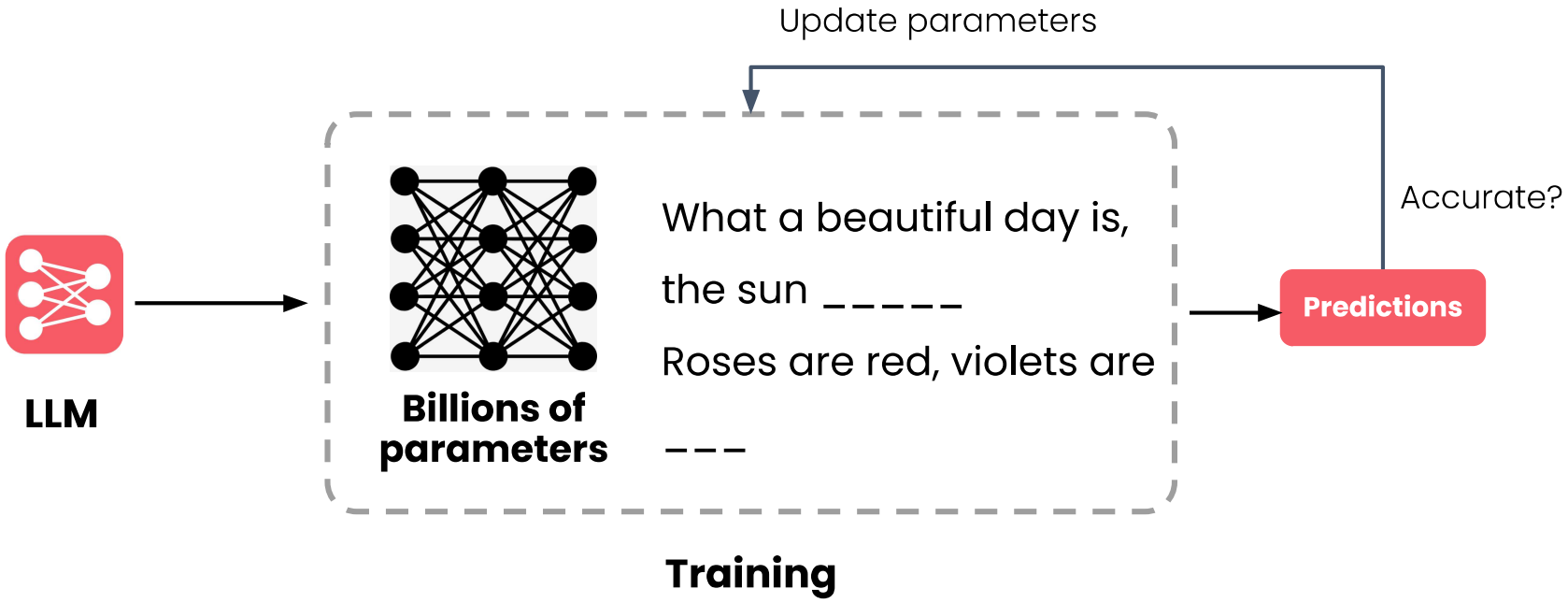





**Before training, LLMs generate gibberish**

"Forward to Saturn’s dance floor!" she yowled, tail transmitting disco beats.

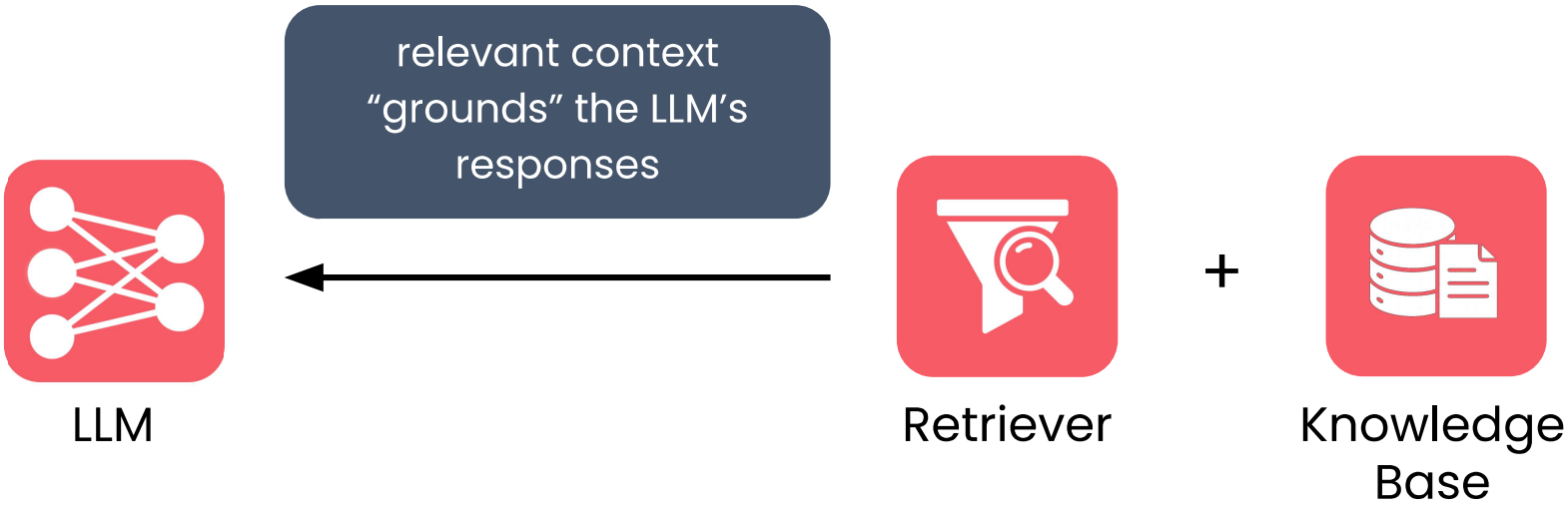
# How LLMs Learn



# Why LLMs Hallucinate

- **LLMs generate probable word sequences**
  LLMs just reproduce statistical patterns from their training data
- **Knowledge gaps cause inaccurate responses**
  Responses can “sound right” but aren't true.
- **Truthful ≠ probable**
  LLMs are designed to generate "probable" text, not truthful text

# How RAG solves the problem



# Why not add everything?

## Higher Computational Cost

- Longer prompts take more computation to run
- Model performs computationally complex scan of every token
- Scan happens before generating each new token

## Context Window Limit

- Eventually you hit the limit of LLM's context window
- Smaller models: only a few thousand tokens
- Largest models: millions

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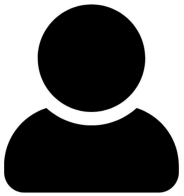
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# Introduction to information retrieval

## Introduction to RAG



**Your Question**



How can I make New York style pizza at home?

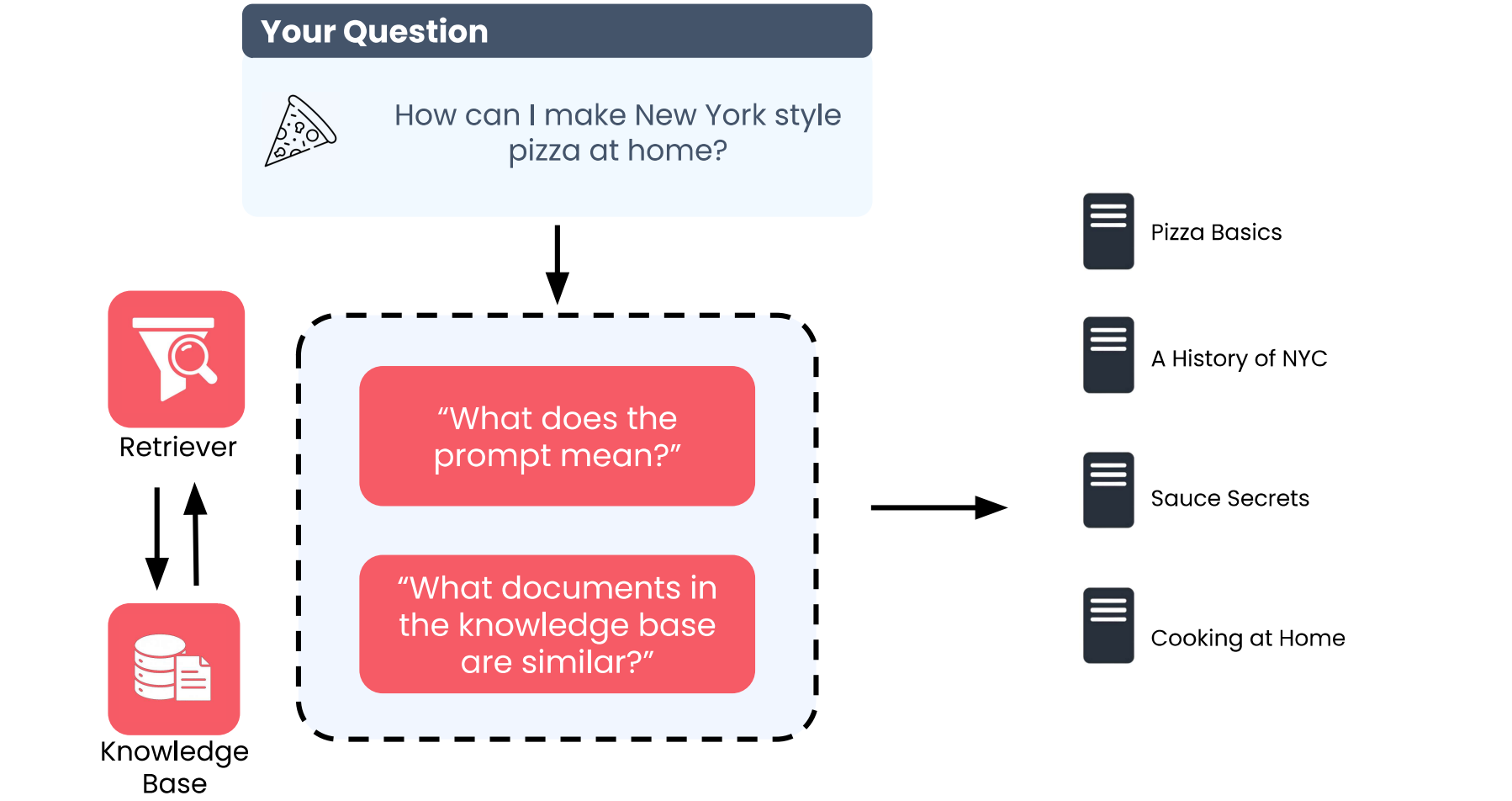
| | Collection | Organization | Search |
|---|---|---|---|
|  **Library** | Books on many topics | Different shelves and sections | Librarian helps you find best sections or books |
|  **Retriever** | Documents in a database | “index” for search | Retriever searches the index |

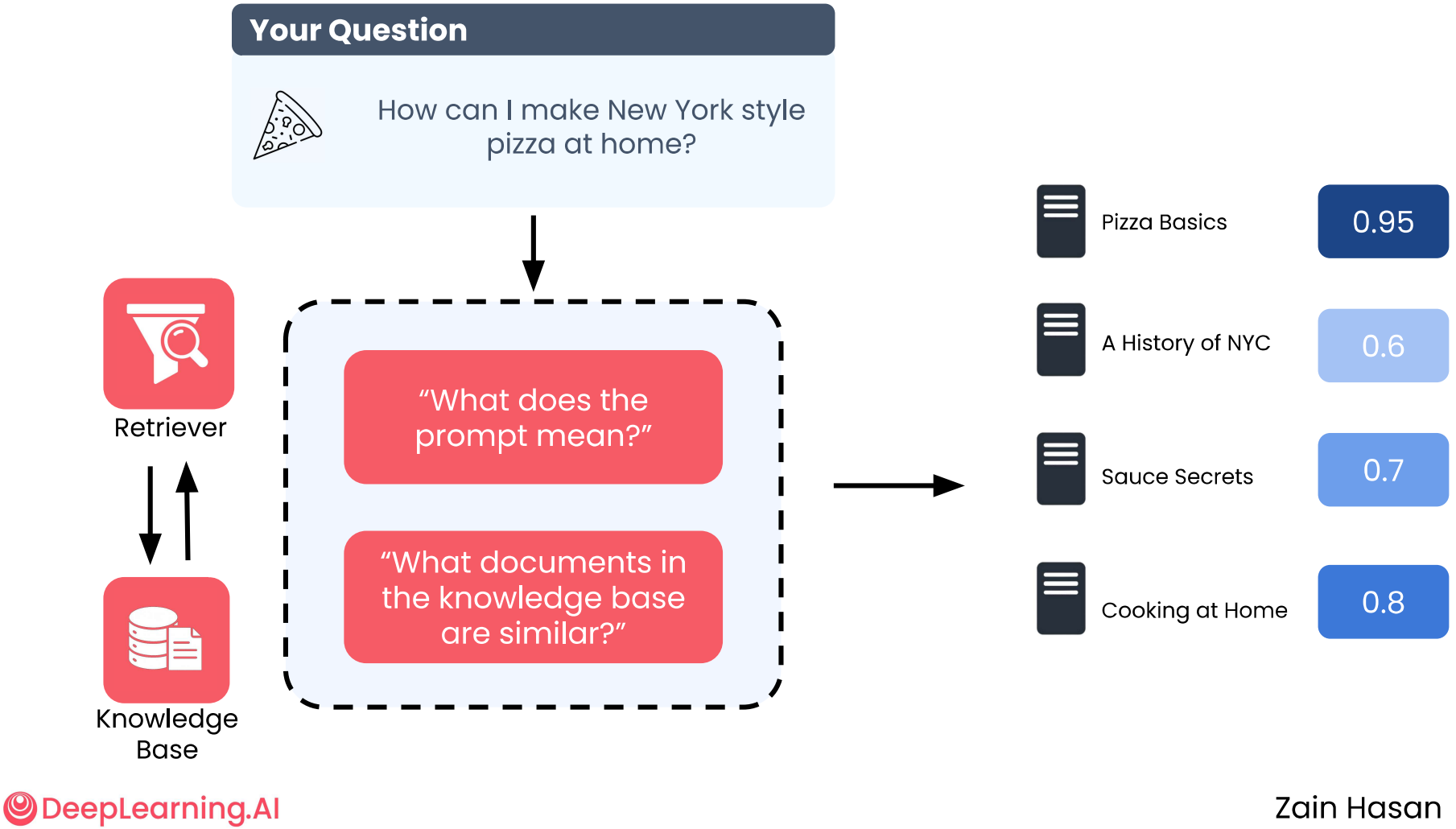
# Search with a librarian



**Librarian**

- Understands the meaning of your question
- Identifies the right shelves to search
- Eventually finds relevant books


Your Question
How can I make New York style pizza at home?
Retriever
Knowledge Base
"What does the prompt mean?"
"What documents in the knowledge base are similar?"
Pizza Basics
A History of NYC
Sauce Secrets
Cooking at Home

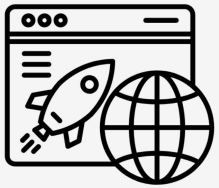

Your Question
How can I make New York style pizza at home?
Retriever
Knowledge Base
"What does the prompt mean?"
"What documents in the knowledge base are similar?"
Pizza Basics
0.95
A History of NYC
0.6
Sauce Secrets
0.7
Cooking at Home
0.8

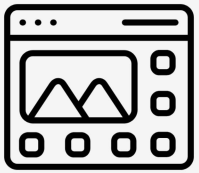
# Retriever Tradeoffs

- **Relevance vs  irrelevance**
  Need to return relevant documents and withhold irrelevant ones
- **Return every document?**
  Mountains of irrelevant docs. Wastes context window
- **Return the single highest ranked document?**
  Miss valuable information
- **No perfect solution**
  Retriever usually doesn't perfectly rank documents
- **Monitor and experiment**
  Change settings to find what works

**Search Engine**

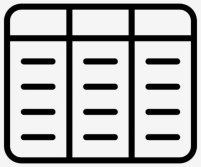
Retrieves relevant webpages





**Database**

Retrieves relevant tables and rows





**Historical Context**

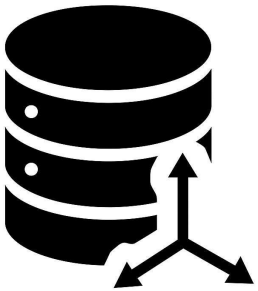
Information Retrieval was already mature when LLMs were first developed

# Practical Implementation



**Relational Database**

Already widely adopted



**Vector Database**

Specialized for retrieval in a RAG system



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# Module 1 Conclusion

## Introduction to RAG

# Key Concepts

- RAG pairs an LLM with a knowledge base
- Data is private, recent, or highly specific and so missing from the LLM's training data
- Retriever finds relevant documents and adds them to an augmented prompt
- LLMs ground their responses in the retrieved information